

Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes three of the following points: Similarities Any two from: • both contain a glycerol (1) • both contain fatty acids (1) • both contain ester bonds (1) Differences • triglycerides have three fatty acids and phospholipids have two fatty acids (1) • triglycerides do not contain a phosphate group but phospholipids do contain a phosphate group (1)	NB Do not piece together from two descriptions in separate sentences IGNORE diagrams	(3)

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8(b)	An explanation that includes the following points: • {protein / phosphate heads / phospholipid heads} are {soluble / hydrophilic / polar} and interact with {blood / plasma} (1) • {fatty acids / triglycerides /cholesterol} is {insoluble / non-polar / hydrophobic} (1) • therefore cholesterol is surrounded by {fatty acid tails / triglycerides} (1)		(3)

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8(c)(i)	- volume of sphere calculated / values substituted into the equation (1) - volume of sphere rounded up to nearest whole number (1) - ratio calculated (1)	Example of calculation = 6912 if using $\pi = 3$ = 7142.4 if using $\pi = 3.1$ = 7234.56 if using $\pi = 3.14$ = 7239.168 if using $\pi = 3.142$ = 7241.472 if using $\pi = 3.143$ = 7238.22947387 if pressing π on calculator = 6912 / 7142 / 7235 / 7239 / 7238 / 7241 ACCEPT 6910 / 7140 / 7240 NB Just these values given = 2 marks 14 : 1 13 : 1 if 6910 / 6912 CE apply throughout NB mark answer in table if different from in the working e.g. <table border="1"> <thead> <tr> <th>Diameter of LDL / nm</th><th>Volume of LDL / nm³</th><th>Volume of cholesterol / nm³</th><th>Ratio of LDL volume to cholesterol volume</th></tr> </thead> <tbody> <tr> <td></td><td></td><td></td><td></td></tr> <tr> <td></td><td>7235</td><td></td><td>14 : 1</td></tr> <tr> <td></td><td>= 2 marks</td><td></td><td>= 1 mark</td></tr> </tbody> </table>	Diameter of LDL / nm	Volume of LDL / nm ³	Volume of cholesterol / nm ³	Ratio of LDL volume to cholesterol volume						7235		14 : 1		= 2 marks		= 1 mark	(3)
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*8(c)(ii)	Indicative content: • as LDL increases, risk increases (K / G) • several factors beside LDLs can increase the risk of CVD (K) • example of a factor given e.g. high blood pressure (K) • LDLs can be different sizes (Q) • and therefore be absorbed by endothelial cells differently (Q) • and therefore get broken down at different rates (K / Q) • and therefore carry different volumes of cholesterol (Q) • level of HDL (in blood) affects risk (of CVD) (K / G) • example given from graph e.g. 0.65 a.u. has greater risk than 2.20 a.u. (G) • ratio of LDL : HDL affects risk (of developing CVD) (K / G) • the lower LDL : HDL the ratio the lower risk of CVD (K / G)		(6)
Own knowledge (K), information given in the graph (G), information in the question (Q) Level 1 : uses either (K), (G) or (Q) 1 mark = 1 comment, 2 marks = 2 comments Level 2 : uses two from (K), (G) or (Q) 3 marks = 3 comments, 4 marks = 4 comments Level 3 : uses (K), (G) and (Q) 5 marks = 5 comments, 6 marks = 6 comments			